VOLLEY | Orbital analysis

Native solver evidence supplement — 18 October 2026

Purpose: show what commanded relative release inputs change in the orbit, how the numerical implementations compare, and what the archived twelve-orbit campaign does not establish.

Case definition

450 km circular reference; 51.6° inclination; 4 kg payload and 300 kg remaining host. At 16.029 m/s relative release, the payload inertial velocity increment is 15.818092105 m/s. The numerical release inputs are prescribed, not measured motor performance.

Execution record

Orekit 13.1.6: queued single-release case, plus eight 24-hour central-gravity/J2 histories, repeated with 60 s and 30 s maximum integration steps. GMAT R2026a: repaired queued case plus two fresh one-day histories. Archived GMAT R1: twelve preloaded post-release orbits, 90 days; not a coupled deployment campaign.

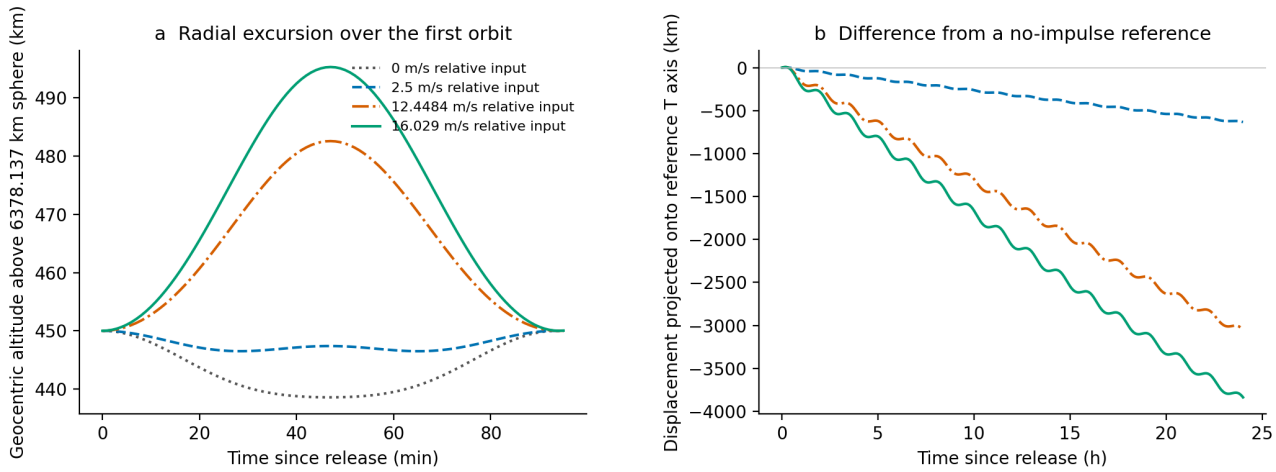
Observed numerical differences

Largest Orekit refinement difference: 6.12996e-06 m. Independent SciPy DOP853 difference: 1.07433e-05 m. Harmonized-central-constant GMAT comparison: 0.227043 m. J2 setup difference: 1.4990 km; not a matched-force validation.

Interpretation

No new hardware qualification, provider compatibility, release accuracy, atmospheric lifetime or collision-risk result is established. Thresholds were not preregistered before exploratory runs. The unresolved J2 setup discrepancy remains a follow-up task. Source, CSV histories, manifests and a detailed reader guide accompany this report.

1 | Release-state consequences



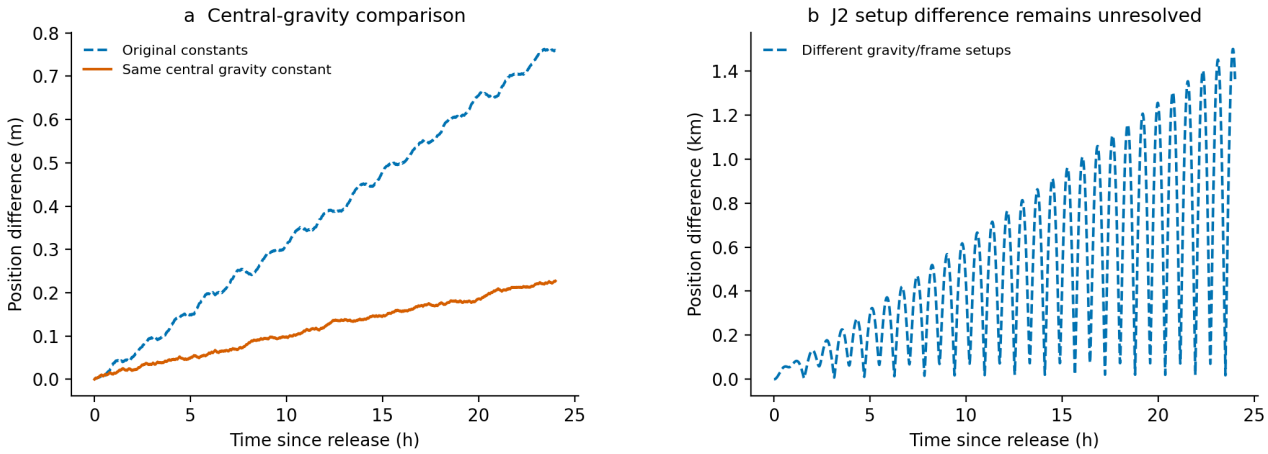
Orekit 13.1.6, fixed-axis J2; 4 kg payload and 300 kg remaining host. Inputs are prescribed, not measured speeds.
T is the reference local along-track axis; this is not a delivery-error budget or a completed mission campaign.

The left panel shows radial excursion; the right is a reference local T-axis projection over one day. The reference axis moves with the no-impulse orbit. Neither panel represents a customer delivery tolerance.

Source and reproducibility

See validation/orbit_evidence/README.md, ORBIT_RESULTS.json and the accompanying raw histories. Figures are Matplotlib exports of computed native output; they are not application screenshots or flight observations.

2 | Solver agreement and setup differences



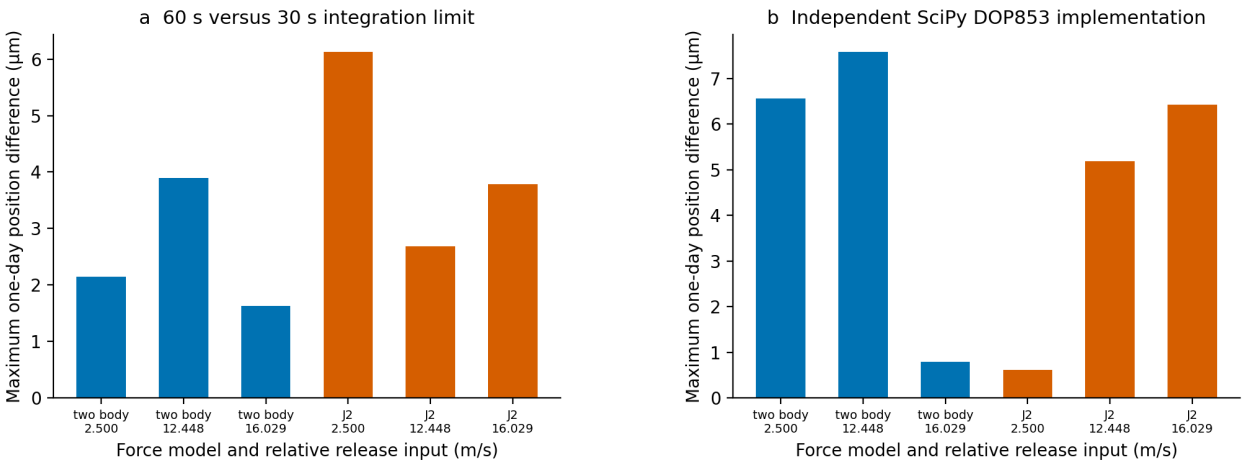
Native GMAT R2026a versus Orekit 13.1.6; identical prescribed initial Cartesian state, 16.029 m/s relative input.
Elapsed output epochs are aligned. The J2 comparison is not a matched-force validation; its residual must not be hidden.

Central gravity constants and output epochs must be reconciled before interpreting a residual. GMAT Earth-fixed degree-2 gravity and fixed-axis Orekit J2 are not identical configurations. The larger J2 discrepancy is retained explicitly.

Source and reproducibility

See validation/orbit_evidence/README.md, ORBIT_RESULTS.json and the accompanying raw histories. Figures are Matplotlib exports of computed native output; they are not application screenshots or flight observations.

3 | Numerical sensitivity



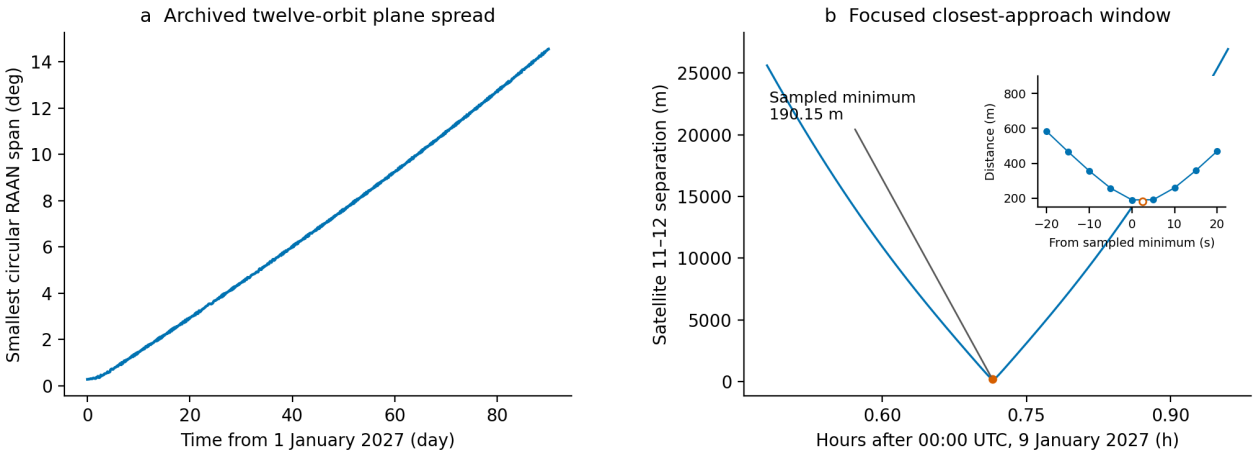
Same force equations and constants; eight Orekit histories including no-impulse cases. These are numerical checks.
Small integration residuals do not bound release dispersion, atmosphere error, host navigation error or flight performance.

Step refinement and a separate implementation test integration consistency under shared physics. They cannot quantify model error from atmosphere, payload contact, control or release dispersion.

Source and reproducibility

See validation/orbit_evidence/README.md, ORBIT_RESULTS.json and the accompanying raw histories. Figures are Matplotlib exports of computed native output; they are not application screenshots or flight observations.

4 | Archived campaign and close approach



GMAT R2026a archived R1: preloaded post-release orbits, not a coupled host campaign. The left plot is downsampled.
The 190.15 m sampled minimum and 180.29 m linear interpolation do not establish continuous clearance or collision probability.

The RAAN span uses circular angular extent. The five-second close-approach sample is 190.15 m; the hollow inset marker is a 180.29 m straight-line interpolation. There is no demonstrated continuous-clearance bound.

Source and reproducibility

See validation/orbit_evidence/README.md, ORBIT_RESULTS.json and the accompanying raw histories. Figures are Matplotlib exports of computed native output; they are not application screenshots or flight observations.